

Lecture 15

Rotor loads, Part 1

ASDA/Dynamics of Rotor Systems
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This lecture

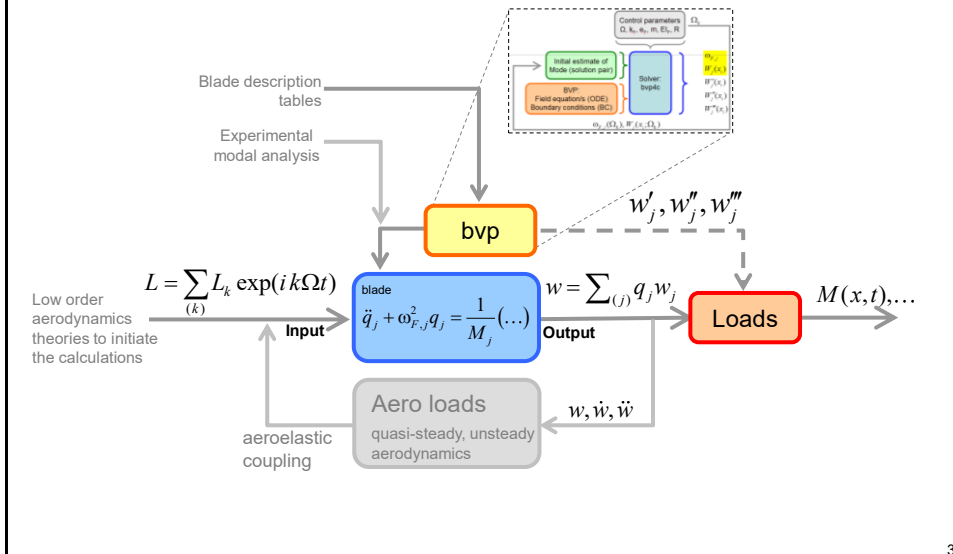
- Blade load calculation methods
- Example: blade spin-up analysis
- Rotor loads terminology

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Basic framework for load calculations

Rotor loads prediction framework developed so far:



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Flapping load calculation methods

Direct differentiation: a *differential method* – uses high mode shape derivatives

$$M(x, t) = EI_F w''(x, t)$$

Requires higher order mode shape derivatives which are provided by bvp4c. Often, however, these quantities are not available. Hence, alternative methods are to be considered (see below).

Modal summation: a *modal method* – uses eigenvalue BVP (see BVP section)

Consider a differential blade element equilibrium: $\frac{dM}{dx} + S - T \frac{dw}{dx} = 0, \quad \frac{dS}{dx} = m\ddot{w} - L$

$$M_i = \int_x^R (-S_i + T w'_i) d\tilde{x}, \quad S_i = \int_{\tilde{x}}^R (-\omega_{F,i}^2 m w_i - 0) d\tilde{x} \rightarrow M = \sum_{j=1}^{N_M} q_j(t) M_j(x)$$

Computationally efficient method based on the previous theoretical developments whilst actively trying to reduce the differential order of the used modal quantities. Still, derivatives can cause problems in the presence of point load effects such as transitions, masses, pitch links, etc.

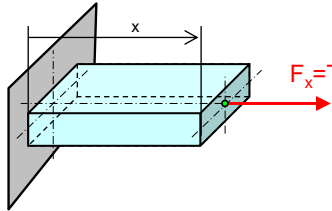
Force integration: an *integral method* - uses load equilibrium on a blade segment

$$M = \int_x^R ((L - m\ddot{w})(\tilde{x} - x) - \Omega^2 m \tilde{x} (w(\tilde{x}) - w(x))) d\tilde{x}$$

Method suitable for blades with "point" loads; lag damper, pitch links, etc. since, by using integration, instead of differentiation, it bypasses the need to represent these discontinuities using a finite set of smooth mode shapes.

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Overview of internal loads using the differential method



Tension due to CF effects:

$$T(x) = \Omega^2 \int_x^R m(\tilde{x}) \tilde{x} d\tilde{x}$$

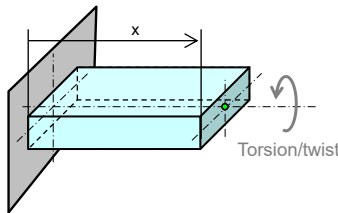
Steady tension due to centrifugal effects.

Axial motion:

Note: axial dynamics was not covered.

$$(EAu')' = \rho A \ddot{u} - f_{a,e}(x,t)$$

(example case of nonrotating bar/blade)



Torsional motion:

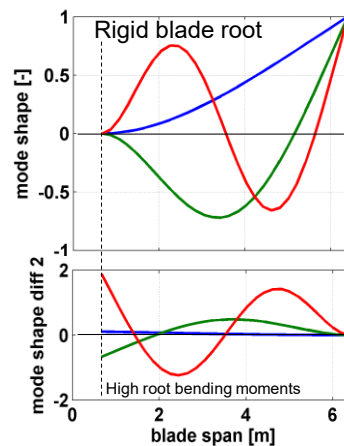
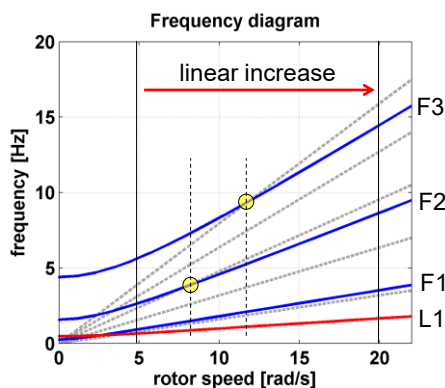
Note: torsional dynamics covered for non-rotating blade

$$(GJ\theta')' = \rho J \ddot{\theta} - t_{t,e}(x,t)$$

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Computational example: blade spin-up analysis

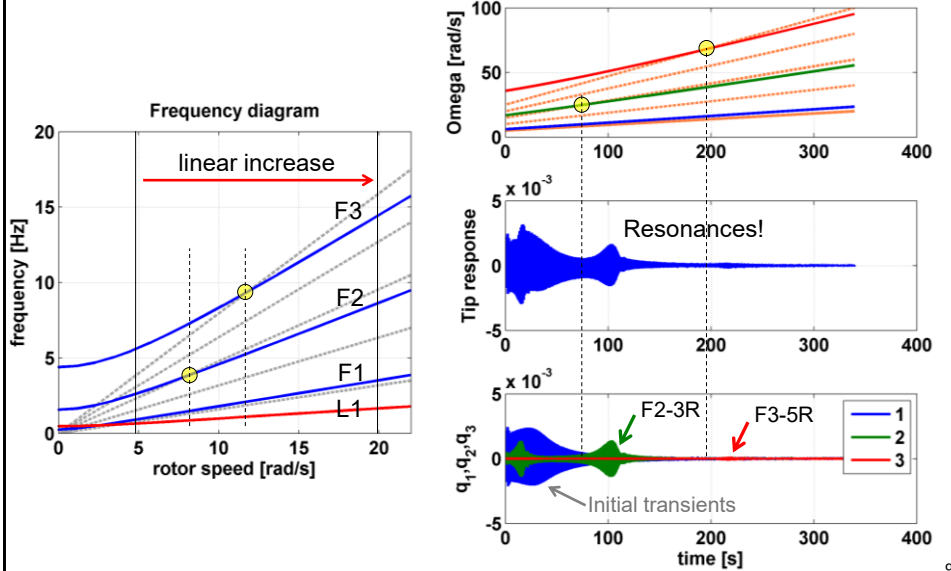
Assume a blade where $EI=5469 \text{ Nm}^2$, $m=4.3 \text{ kg/m}$ and $R=5 \text{ m}$. Consider the first three flapping modes and Ω changing from 5 to 20 rad/s in 4.5 minutes. Study blade flapping response and blade root moment. First five higher harmonics of the aerodynamic loads ($R1, \dots, R5$) are approximated by arbitrary distributions on the blade.



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Computational example: blade spin-up analysis

Form three equations for the mode participation factors q_1, q_2, q_3 . Solve in MATLAB using an ODE solver and synthesize the flapping response.



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Moving from blade to rotor loads

Three systems of loads are usually considered during analysis:

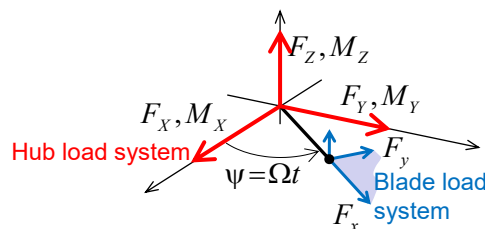
- rotating **blade loads**,
- rotating hub loads – sum of all blade loads in the rotating CS,
- non-rotating **hub loads** – vector transformation rules, see later.

These groups of loads can be distinguished:

- rotor in-plane forces and moments (moment vectors),
- rotor out-of-plane force and moments (moment vectors).

Individual loads can be further classified as:

- steady or constant load components,
- vibratory or harmonic load components.



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Summary

- Differential, modal, force integration blade load calculation methods can be used to find the blade sectional loads.
- Individual blade root loads are transferred to the rotor hub. Other possible load paths, not discussed here, may need to be considered (pitch links, lag dampers, etc.).
- Hub loads represent an aggregate effect of all loads introduced by the individual blades at the points where they meet the hub.
- The vibratory hub loads are the primary source of the dynamic fuselage loading.

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